

Rethinking Participatory Art under Decentralization in Web3

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Abstract

The emergence of Web3 introduces decentralization to online spaces and applications. In particular, Web3 participatory art extends this concept to artistic creation. This late-breaking work provides an initial exploration of the impact of Web3 decentralization on the collective artistic creation of participatory art, and potential strategies that can help artists adapt to this new technological environment. Grounded in smart contract technology, we examine prominent Web3 participatory art projects and discuss existing implementation for achieving decentralization. Building upon this, we conduct a fully decentralized participatory art project with a broad scope of decentralized creative functionality. By analyzing participants' on-chain behavior, off-chain communication, and motivation data, we unveil potential shifts in artists' responsibilities and provide design considerations for decentralized participatory art.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**; • **Applied computing** → **Media arts**.

Keywords

Web3, Participatory Art, Decentralization, Blockchain

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1 Introduction

As people increasingly value democracy and freedom on the internet, the current read-write Web2 dominated by a handful of tech

giants no longer satisfies the increasing desire for a more user-empowered internet [2]. Web3, which embraces the concept of decentralization guaranteed by blockchain technology, has thus been proposed to counter the centralized structure (and its adverse consequences) of the traditional Web2 ecosystem [27, 33, 42]. Specifically, Web3 applications utilize technologies such as smart contracts and non-fungible tokens (NFTs) to reduce the necessity of centralized control. Smart contracts are self-verifying, self-executing and tamper resistant programs on blockchain, which enable the execution of rules without centralized servers [11, 25, 44]. Meanwhile, NFTs, typically managed through smart contracts, are unique assets that can reference content with verifiable creation, transfer, and ownership in a disintermediated manner [5, 32, 43]. Web3 and its associated technological innovations are increasingly regarded as having the capacity to offer an emerging decentralized online space, with the potential to reshape the roles associated with identities and, consequently, related user behavior patterns [1, 6, 13, 14, 16, 37]. Crypto art, with NFTs serving as the medium for artworks, has become one of the most significant application domains born out of Web3 [7]. However, the success of blockchain technology in the cryptocurrency domain [26] has endowed Web3 with intrinsic economic properties [42], significantly influencing research on crypto art to focus heavily on its economic aspect [28, 40, 45].

Although researchers have pointed out the significant involvement of NFT buyers in the NFT creation process [14], the impact of Web3 decentralization on the creative power, traditionally centralized in the hands of the artists, continues to be overlooked by academia. Interestingly, a revitalized and widely recognized art form has emerged within the Web3 industry: participatory art, which encourages public participation in the creative process of artworks [10]. By leveraging the power of Web3 to overcome the limitations of traditional participatory art, this art form has the potential to extend decentralization from market circulation to artistic creation. In traditional offline and Web2 contexts, the tools and platforms (physical venues and online websites) that facilitate and host the mechanisms of participatory art are managed by centralized institutions or individuals. As initiators, artists inherently possess centralized power relative to the participants, having the ability to modify the creation mechanism at any time, directly exerting control over the creative behavior of participants, and even denying participants collaborator status following the artistic co-creation [10, 12, 22]. By contrast, in Web3, smart contracts

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have the capability to provide participants, who are usually NFT buyers, with immutable mechanisms and decentralized execution that cannot be interfered with by artists or other centralized organizations [25]. In addition, NFTs can offer participants trustless recognition for their contributions to the creative process through verifiable records and ownership [43]. With this new technological support, participants have the opportunity to possess truly equal creative power with artists under the mechanisms of participatory art, rather than solely relying on that voluntarily granted by artists. While inherently challenging to achieve in traditional platforms, decentralization in artistic creations has begun to germinate in the Web3 ecosystem.

To explore the impact of decentralization on the artistic creation of participatory art, it is necessary to develop a highly decentralized art project to provide a relatively pure decentralized environment and comprehensive data. However, current Web3 participatory art lacks research on the implementation of decentralization. In this late-breaking work, we investigate seven well-known Web3 participatory art projects, thus summarizing current existing decentralization methods. Building upon this groundwork, we design and conduct a fully decentralized participatory art project with a broad decentralization scope of creative functionality. We analyze the off-chain chat records using the Co-Occurrence Network. Additionally, by examining the distribution of motivations corresponding to on-chain creative behaviors on the canvas and timeline, we discuss the tendencies of different motivations in decentralized artistic creation. Simultaneously, we employ the chi-square test to assess the correlation between creative behaviors and motivations. Based on the results of the experimental project, we argue that in decentralized participatory art, the responsibilities of artists should shift. Specifically, rather than solely expressing personal views, artists need to heavily focus on determining the decentralization scope of the creative functionalities, and employing diverse methods to motivate participants to achieve the intended creative goals. Moreover, artists should engage in off-chain operations to maintain participant community vibrancy. This paper calls for further investigation into the potential impact of decentralization on artistic creation, an area that has been overlooked in current research and practice.

2 Method

2.1 Decentralization Implementations

Although the inherent characteristics of blockchain technology and the ethos of Web3 community both revolve around decentralization [27, 29], there remain uncertainties about how participatory art can be decentralized in Web3. For many Web3 participatory art projects, decentralization appears to be more of a gimmick to align with Web3 rather than a genuine spirit of artistic creation, in the sense that they do not truly incorporate decentralization in their technical implementations.

As Web3 decentralization is fundamentally built upon blockchain technology [29], the influence of decentralization on participatory art is predicated on the on-chain deployment of all or part of its mechanisms through smart contracts. To explore the current implementation methods of decentralization, we conduct an in-depth investigation of seven well-known on-chain participatory art

projects, including Merge [18], Your Daytime Fireworks [39], LOST-POETS [30], First Supper [15], The Space [38], Hashmasks [19], and Entrope [4]. The results are summarized and organized in the following six aspects related to the integration and application of smart contracts:

1. Participatory function implementation. The immutability of blockchain [47] guarantees the fixity of smart contract code [11]. Thus, it is crucial to establish consistency between the participatory mechanisms defined by the artists and the participatory functions implemented in the smart contracts, ensuring the immutability of the mechanisms. Contract-based functions should run autonomously on-chain without reliance on off-chain computation, precluding opaque, centralized interference. Additionally, these contract functions should be encoded in a single contract or a series of immutable, interconnected contracts to prevent artists from fundamentally changing the predefined participation logic by replacing contracts.

2. NFT metadata and media content handling. NFT metadata describes crypto artworks and provides references to their specific media content [17, 43]. NFT-related content should be automatically allocated and updated within the contract, without allowing individuals to have special privileges to replace it. Meanwhile, the files pointed to by the URIs should be stored on InterPlanetary File System (IPFS) [8], a distributed storage system widely used in the NFT field [43], to ensure the immutability and permanent accessibility of the information. Moreover, when generating the metadata of NFTs in contracts, issues similar to those encountered during the implementation of participatory functions arise. Specifically, if NFT metadata generation logic is split into replaceable smart contracts, artists can modify this logic by replacing these contracts.

3. Artist control during the creation stage. In smart contracts, addresses with special permissions can be set to call specific functions. When contracts do not set administrator privileges in the participatory functions and NFT generation functions beyond those of regular participants, artists lose the ability to directly interfere with the participants' creative behavior.

4. Participant selection. Due to the pseudonymous nature of blockchain [46], the entities interacting with smart contracts in participatory art are cryptographic addresses that do not reveal the real-world identities of participants. Unless artists can obtain the off-chain information of participants and manually configure a participant list within the contract, anyone can participate in the artistic creation by following the predefined rules. Consequently, the power to select participants becomes decentralized.

5. Proxy contract utilization. Although the code in smart contracts is immutable, the most common method employed by current Web3 applications to ensure project updatability is the simultaneous deployment of proxy contracts and logic contracts. Contract creators can modify project mechanisms by replacing the logic contract while retaining the proxy contract [36]. If participatory art projects employ proxy contracts, all mechanisms can technically be modified in a centralized manner by the artists.

6. Smart contract accessibility. The transparency of blockchain technology allows anyone to inspect the code and transactions of smart contracts, thereby establishing trust and enabling the monitoring of behaviors [34]. In participatory art, smart contracts' accessibility is a prerequisite for the advantages brought about by

transparency. This is achieved by providing contract addresses and verifying contracts into human-readable formats, which motivate artists to refrain from engaging in centralized actions that violate consensus.

Although blockchain technology introduces the concept of decentralization to participatory art creation, this concept cannot extend to the entire artistic process. During our investigation, we observed that Web3 participatory art is divided into two creative stages, with artists and participants leading each stage respectively. This situation aligns with the definition of participatory art as a form of art creation that involves the public in the creative process [10], seeking a balance of power between participants and artists rather than completely eliminating the need for artists [22]. In the first stage, the artists bear the responsibility of designing the creative mechanism and determining the scope of decentralization of the power. They then develop the decentralized mechanism through smart contracts. Once rights are decentralized, the artist relinquishes complete control over the relevant rights. The second stage commences with the deployment of the smart contract. Based on the decentralization scope, the deployed smart contract may supersede the artist's dominance, becoming the supervisor and executor of the project's mechanism. Participants can freely exercise the decentralized creative rights within the framework set by the artist. Therefore, even in the case of complete decentralization, decentralization should occur in the second stage.

2.2 Experimental Project

Since the existing participatory art projects have to consider various factors in their actual implementation, such as high gas fees and long execution times caused by complex on-chain mechanisms, the scope of their decentralization is limited from a technical perspective. To investigate the impact of decentralization on collective artistic creation, we developed a fully decentralized experimental project with a broad scope of decentralized creative functionality using the Polygon mainnet. The developed project provides an uncontaminated environment for decentralized creation and enables the acquisition of complete on-chain and off-chain information for analysis. This project embraces decentralization as its core tenet, granting participants absolute autonomy within the creative mechanism. To ensure complete decentralization, we strictly adhere to the implementation methods discussed in the previous subsection. Moreover, we adopt pixel co-creation as the primary mode of expression, empowering participants with extensive creative freedom.

Participants can determine the co-creation target. Towards this end, they are given one week to independently create artwork that expresses their personal ideas using the specified colors on a canvas of the corresponding size provided through the pre-creation webpage. They can then post their artwork to the designated Discord channel to freely discuss and vote for the most suitable artwork as the collaborative goal. After one week, the artwork with the highest number of votes becomes the co-creation target. With the target decided, co-creation begins with red and blue initial participants, with the red paint permission being the target author and the blue one randomly selected from pre-creation authors. Upon being granted painting permission, a participant acquires a specific color to imbue any empty pixel, thereby minting an NFT associated

with the pixel's coordinates and color. However, this permission is ephemeral and is lost after two painting actions, but can be regained by transferring one's current color to a new participant. The original participant's new color will have 20% reduced saturation, while the transferred participant receives the original color without saturation reduction to commence their artistic contribution. This process can be continuously repeated. Once the minimum saturation threshold is reached, subsequent transfers yield unaltered colors. Participants collectively control new member selection and color rarity. The evolving canvas can be minted as a dynamic NFT, with sale proceeds distributed proportionally to participants' pixel NFT holdings. All creation behaviors have matching functions in the smart contract, and each behavior step triggers an on-chain transaction. Concurrently, to further explore the profound effects of decentralization, we collect motivation data behind every on-chain behavior instance.

Although the field of participatory art has not comprehensively explored motivations of participants, we can gain insights from related fields. Motivations are typically divided into two categories: intrinsic and extrinsic [35]. Studies on Web2 participatory art suggest fun as a primary intrinsic motivation for participation [23]. For NFT, which serves as a vital medium of Web3 participatory art, related research indicates three distinct categories of economic motivations among NFT buyers: short-term profit speculators, long-term profit investors, and long-term holding collectors [3].

Moreover, investigations within the NFT domain underscore the profound connection between community and NFTs [21], resonating with the notion that community plays a pivotal role in Web2 participatory art [22]. This confluence of findings intimates that community is likely the potential motivation for decentralized participatory art. To consider the potential impact of community, we explore existing research on online crowdsourcing platforms, which is a domain entailing similar collaborative creative participation. Among the social motivations proposed by Mokter Hossain [20], we screen out the motivations exhibiting relevance to artistic creation, including the intrinsic motivation of community drives and the extrinsic motivations of collaboration and reputation. Additionally, the decentralized nature confers autonomy on participants, potentially enabling a competitive behavior that is contrary to the collaborative essence of participatory art. Thus, gaming motivation may manifest as a potential driver. Although intrinsic motivation is considered more related to creative conduct in traditional creative fields [41], a contrasting phenomenon is observed in the current Web3, where economic motivation emerges as a crucial component underlying engagement with NFT-related activities [3, 31].

To explore the manifestation of motivations in decentralized participatory art while emphasizing the distinct influence of economic motivations under the Web3 context, we collect motivations separately according to economic and non-economic categories. When participants perform each on-chain creation step, they first have to self-evaluate their current motivation before executing the associated contract function.

3 Results

The fully on-chain creation functions enable participants to control the duration of the creative process, potentially allowing it, from a

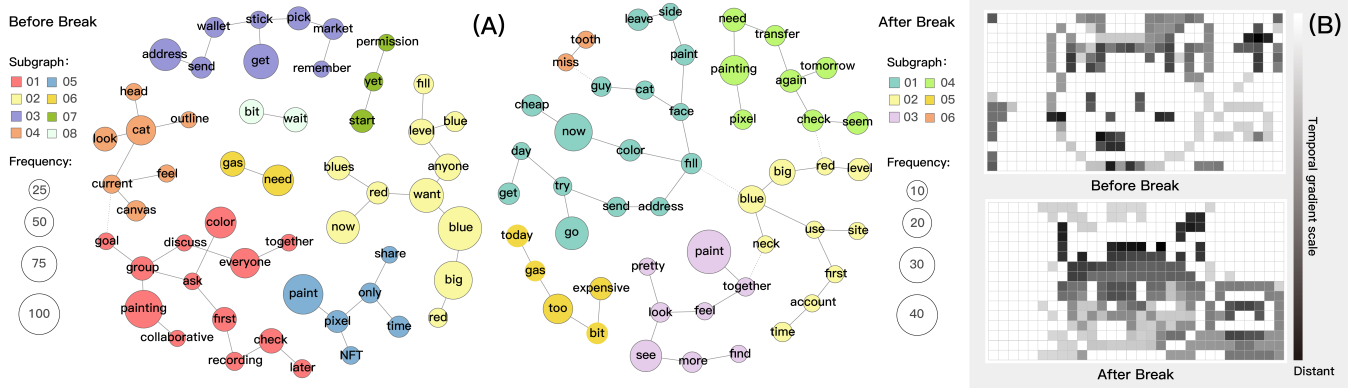


Figure 1: (A) is the Co-Occurrence Network of chat records before (left) and after (right) the break. Nodes of the same color refer to the same topic. (B) depicts the temporal distribution of pixel placement on the canvas before (upper part) and after (lower part) the break.

technical perspective, to continue perpetually. However, due to the absence of any new information appearing on either the blockchain or the group chat for the past four months, we deem the project to have concluded. By considering the initial and final instances of on-chain and off-chain information as reference points, the project lasted a total of 114 days. During the project, 77 addresses participated in the co-creation, drawing and minting 354 pixel NFTs. However, due to the inherent pseudonymity of blockchain technology, it is challenging to confirm whether these addresses represent unique individuals. The possibility that some participants may have controlled multiple addresses cannot be ruled out. The number of on-chain interactions reached 480, while off-chain communications amounted to 2,241 messages.

During the period of time dedicated to determining the co-creation target, participants submitted 25 candidate targets. Among them, a portrait of a blue cat with a red heart shape in the upper right corner was selected as the final target. While the target was determined, the collaborative result exhibits numerous detailed modifications made during the creation process, including a more rounded cat head outline, the addition of a portrait in the lower right corner, and the preservation of blank spaces. During the project operation, we maintained decentralization by refraining from interfering with participants' discussion topics, except for necessary rule explanations and publicity. We observed that the project halted on November 14th, after 43 days of continuous operation, with the co-creation target yet to be achieved. After waiting for 39 days and finding no signs of participants spontaneously restarting the project, we intervened in the off-chain chat group, urging participants to continue creating, which allowed the project to resume. A strict break, defined as no data being collected off-chain or on-chain, occurred between November 14th and December 7th. Although some chatting and on-chain interaction occurred on December 7th, most communication and creation began after our call for continued creation on December 23rd.

To investigate the differences in chat topics before and after the break, we conducted a Co-Occurrence Network topic analysis on the off-chain chat records, following the strict break differentiation. The results are shown in Figure 1 (A). The chat topics before the

break (left side) were more complex, involving question-oriented subjects such as wallet settings (color 3), permissions (color 7), and mechanisms introduction (color 5). Regarding creation-related topics, the before-break phase focused more on discussions that promoted group creativity to reach consensus (color 1) while concentrating on key features in the painting (color 4), such as outlines and the head. The after-break topics (right side) mainly revolve around labor planning in chronological order, such as now (color 1) and tomorrow (color 4) tasks. Additionally, there are evaluations of the final creation results (color 3) and a small number of discussions on painting details (color 6). Regardless of the break, the color transfer mechanism remains the second most discussed topic (color 2), indicating that our mechanism design indeed provides opportunities for social interaction. Overall, before the break, participants focused on creative expression, utilizing group creativity to achieve creation goals and the main features of the painting; after the break, participants were more engaged in repetitive labor to fill in the picture. The different on-chain painting positions before and after the break, shown in Figure 1 (B), confirm this result. Before the break (upper part), participants drew the main body of the painting, such as the cat's facial features, the complete outline, and other key features, as well as part of the portrait's head. After the break (lower part), participants focused on coloring the picture's content.

Additionally, although color 6 before the break and color 5 after the break both involve on-chain gas consumption, pre-break communication discusses the project's fully on-chain nature that requires gas to execute on-chain operations, while post-break discussions focus on whether the current gas fees are expensive. This characteristic also extends to the now task (color 1) topic, where participants await low gas fees to paint. This phenomenon is reflected in the on-chain data: before the break, the gas fee consumed for painting is ($M=0.04$, $SD=0.01$), and transferring color is ($M=0.007$, $SD=0.003$); after the break, these consumptions decrease, with the gas fee consumed for painting being ($M=0.02$, $SD=0.02$) and transferring color being ($M=0.004$, $SD=0.002$). We speculate that creative behavior leads participants to ignore gas consumption, while repetitive labor heightens mindful consumption, leading to cost-conscious on-chain interactions.

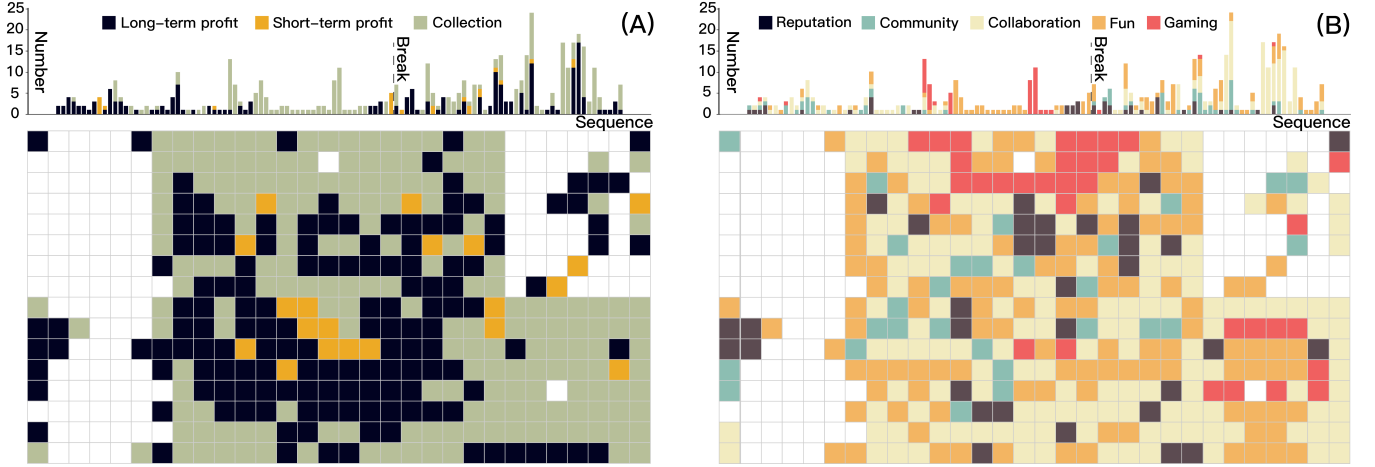


Figure 2: Economic (A) and non-economic (B) motivations behind on-chain creative behaviors. The stacked histograms depict the sequence of motivations (x-axis) against the number of motivations occurring within half an hour after a motivation emerges (y-axis). Blank intervals exceeding half an hour between motivation occurrences are omitted. The lower part illustrates the spatial distribution of motivations on the canvas.

We collected the motivations behind each on-chain behavior, plotted them as a stacked histogram according to their order of appearance, and placed the motivations associated with the painting function at the corresponding positions on the canvas to observe their impact, as shown in Figure 2.

The results show that economic motivations exhibit a cyclical distribution before and after the break: the early and middle stages of each cycle are largely dominated by long-term profits; the end of each cycle is dominated by NFT collection motivation. The distribution of motivations is clearer before the break and more complex after the break. Moreover, the mapping of economic motivations on the canvas indicates that key features, such as cat outlines and character outlines, are mainly driven by NFT collection motives, where long-term profit motives dominate the coloring of the cat's head, and short-term profit motives tend to choose the center of the artwork for drawing.

The cyclical distribution of non-economic motivations is not obvious, but before the break (creative stage), the proportion of fun and gaming motives is higher, while after the break (repetitive labor stage), collaboration motives dominate. The distribution of non-economic motivations on the canvas shows that the key features of the painting are mainly completed by fun motives, while collaboration motives are more inclined to mechanical coloring. By combining the co-creation results and the temporal order of pixel placement, we found that gaming motives may be the potential trigger for new features in the painting. For example, the drawing of the portrait began with gaming motives, and subsequent participants then started painting.

To investigate which motivations are more readily elicited by the main creation functions, namely painting and transferring color, we conducted a chi-square test to examine the correlation between these categorical variables. Given the distinct motivation distributions before and after the break, the test was performed separately for each period. Notable findings include: Before the break, the

"transfer color" function significantly correlated with non-economic motivations ($p=0.040$, $p<0.05$). Additionally, the standardized residual with an absolute value greater than 1.96 was community motivation (2.437), suggesting that this function is more likely to attract community motives. After the break, the "painting" function significantly correlated with non-economic motivations ($p=0.009$, $p<0.05$), with a reduced likelihood of attracting community motives (standardized residual=-3.094).

4 Discussion

In this section, we first briefly overview and compare the different participatory art models. Based on the data analysis presented in the previous section, we then discuss the potential impact of decentralization on participatory art, and propose possible countermeasures, if and when needed.

Traditional offline participatory art emerged in the 1960s with the objective of empowering individuals or collectives with creative agency, making the art process more democratic and egalitarian, and restoring lost social bonds [9]. The advent of Web2 had a significant impact on participatory art, leading to transformative changes such as expanded avenues for participation, increased flexibility in engagement times, innovative online creative methods, and more deliberate participation motivations [23, 24]. This online participatory art is defined as crowdsourced art. Although the openness of the creative process blurs the boundaries between artists, audiences, and artworks, and as already mentioned in Section 1, the artists, as initiators, still maintain in both traditional offline as well as online participatory art projects centralized power throughout the entire process, deciding who participates in the creation, whether to interrupt or intervene in the participants' creation, whose contributions will be retained, and how the artwork will be documented and presented to the off-site audience [10]. In both models, participants can contribute free labor following the artist's instructions, and often struggle to attain co-authorship status [22]. In the emerging Web3,

participatory art has become increasingly popular due to its decentralization of creative rights and empowerment of participants, which aligns well with the decentralization ethos of Web3.

Decentralization fosters parity between artists and participants, granting participants the possibility to engage in artistic creation without interference from the artist. Participants can freely express themselves within the participatory art mechanism safeguarded by the smart contract, and obtain tamper-proof validation of their creation records guaranteed by the blockchain. Participants are the direct beneficiaries of decentralization. At the same time, artists also benefit from the unpredictability introduced by the participants' free behavior, which expands the unknowns and intrigue of the artwork. However, from the perspective of artistic realization, the unpredictable behavior of participants may also engender uncontrollable outcomes. The broad range decentralization of the above experimental project has shown that the final manifestation of the artwork can be entirely unpredictable throughout the project duration, which challenged the traditional mindset of artists to control the overall situation. Once a creation function is decentralized on-chain, artists will completely lose control over it. Therefore, it becomes necessary for artists to exercise judiciousness in determining which aspects of the creative mechanism are suitable for collective realization. Otherwise, broad scope decentralized creation may get out of control. For example, a small group of participants disrupted the original heart icon in the upper-right corner of the co-creation target, resulting in visual chaos in that region of the final work.

In addition to the careful consideration of the decentralization scope, we assert that the strategic selection of participants through mechanism design and publicity could be pivotal for ensuring the quality of decentralized participatory artworks. While decentralization elevates the standing of participants, granting them greater creative agency, the pseudonymity afforded by blockchain technology poses a significant challenge for artists in directly selecting participants. This renders the leaders (participants) of the artistic creation in the co-creation stage highly unpredictable: participants may range from amiable collaborators to rude disruptors. The conducted data analysis reveals that participants with varying motivations exhibit distinct inclinations toward the creation content and methods in decentralized environments. Therefore, when designing participatory art, artists could employ creative mechanisms to attract participants driven by specific motivations, thus exerting some partial control over the direction of the creative process. Data from experiments, potentially serving as valuable references for artists and related researchers, indicate that participants motivated by NFT collection and fun are inclined towards creating innovative content, while gaming motivations may introduce novel creative ideas. Conversely, participants driven by long-term profit and collaboration motivations are more likely to undertake repetitive tasks.

Furthermore, within the realm of decentralized participatory art, artists may need to take on organizational and operational responsibilities. As evidenced by the observed break in our experiment, participants autonomously partition the co-creation process into two distinct phases: initial key feature drawing, followed by coloring. After completing each phase, and in the absence of the artist's intervention, participants lost their desire to create, causing the creative process to be interrupted. In decentralized participatory art, participants potentially tend to break down the creative content

into small phases and complete them sequentially. Therefore, while decentralization is the core value in Web3, artists should still play a guiding role off-chain to mobilize participants' enthusiasm and ensure the artwork's final output.

The role of artists in decentralized participatory art appears to be undergoing a transformation, shifting from traditional individual expressionists to mechanism designers, project organizers, and operators. Facilitated by blockchain technology, decentralization grants participants genuine autonomy, therefore transferring the creative power from artists to participants and prompting artists to adapt their responsibilities to accommodate this shift. Moreover, decentralization ensures that participants collaborate in creating artworks with equal status, thus achieving the original aspirations of participatory art in stimulating collective creativity, advocating egalitarianism, and mending the social bond [9].

However, Web3 is still in its early stages of development. This leads to a series of common inconveniences for on-chain applications, including slow on-chain processing, gas fees required for every on-chain execution, and the need to set up specialized wallets [33]. These user-unfriendly factors hinder decentralized participatory art from achieving true inclusivity. Traditional Web2 users find it difficult to overcome a range of digital divides to successfully join decentralized participatory art projects. At the current stage, decentralized applications are only directed at Web3 users, rather than the broader Web2 community. With future Web3 technological advancement, it is hoped that focus is put on the development of user-friendly and accessible Web3 technology, thus bridging the current digital divide.

5 Conclusion and Future Work

This late-breaking work explores how the decentralization introduced by Web3 impacts the creation process in participatory art, calling for researchers and practitioners to focus on the potential reshaping of collective artistic creations brought about by decentralization. We investigated seven prominent Web3 participatory art projects and synthesized existing decentralization implementation methods. In order to establish a purely decentralized environment and gather comprehensive on-chain and off-chain data to explore the influence of decentralization, we executed a fully decentralized experimental project on the Polygon mainnet. Subsequently, we meticulously collected and analyzed thorough creation-related data, including chat records, on-chain interactions, and behavior motivation. Finally, we discussed the experimental outcomes and proposed preliminary insights into the impact of decentralization on participatory art, along with potential coping strategies for artists.

Due to the absence of motivation exploration in the decentralized participatory art field, the motivations for on-chain behavior in this exploratory study were derived from related research areas. In future work, we will conduct in-depth interviews to further investigate participant motivations. Additionally, constrained by the operational difficulties of real-world projects, the low economic value of NFTs in the experimental project makes it challenging to fully explore short-term motivational behaviors. Future work will consider collaborating with commercially successful projects that have achieved relatively high levels of decentralization to conduct more in-depth research on the impact of short-term profit incentives

on decentralized creation, and to study the effect of participation financialization on artistic integrity.

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